Authentication Log Investigation

Synthetic authentication dataset • Incident timeline and defensive recommendations • 1 August 2026 (UTC)

Scope: This report analyzes the 12 rows in the supplied *authentication-log-sample.csv*. The IP addresses are documentation-range addresses and the account IDs are fictional. Findings describe patterns in this sample, not a confirmed real-world compromise.

1. Executive summary

The sample contains 12 events across six account IDs and three source IP addresses. Two patterns merit investigation: repeated failures against one account followed by a lock event, and rapid failures against multiple accounts from one source IP followed by a rate-limit block. A later successful login for **stu-311** comes from a different IP and device than the earlier failed attempts against that account; this is a review lead, not proof of account compromise.

Measure	Observed result
Total events	12
Password-login failures	7
Successful events	3 (two password/refresh successes and one account-lock event is not counted as a login success)
Blocked events	1 rate-limit block
Distinct accounts	6: stu-104, stu-117, stu-209, stu-311, stu-415, stu-502
Distinct source IPs	3: 192.0.2.18, 198.51.100.27, 203.0.113.42

Priority: Preserve relevant authentication and identity-provider logs; review the lock and rate-limit policy; validate the later stu-311 login against expected user/device activity; and check whether the sample omits related events.

2. Data normalization

The supplied file already has consistent column names and ISO-8601 UTC timestamps. For analysis, timestamps were parsed as timezone-aware UTC values and events were sorted chronologically. Account IDs, source IPs, event names, result values, and device labels were preserved as recorded; no geolocation or real identity was inferred from documentation-range IPs.

Field	Normalization / interpretation
timestamp	Parse as ISO-8601 UTC; preserve the trailing Z meaning UTC; sort ascending.
user_id	Treat as a fictional account identifier; do not equate it with a person's real identity.
source_ip	Normalize as IPv4 text; group events by exact IP. All three addresses are documentation ranges.
event	Keep event types distinct: password_login, account_lock, refresh_token, rate_limit.
result	Preserve success, failure, and blocked as distinct outcomes; do not count account_lock success as a successful login.
device	Preserve the reported device label, including unknown; device labels alone are not strong identity proof.

3. Detection rules and results

These are practical triage rules for this small dataset. They are heuristics, not universal thresholds; tune them to normal user behavior and the system's authentication policy.

Detection	Suggested rule	Dataset result	Assessment
Brute-force-like	At least 3 failed password logins for one account from one IP within 5 minutes.	stu-117: 3 failures from 198.51.100.27 between 09:03:44 and 09:04:18 UTC (34 seconds).	Triggered. Account lock appears at 09:05:03 UTC.
Password-spraying-like	One IP attempts password login against 4 or more distinct accounts within 2 minutes, especially with unknown device metadata.	203.0.113.42: failures for stu-209, stu-311, stu-415, stu-502 from 09:21:10 to 09:21:22 UTC (12 seconds).	Triggered. Pattern is consistent with spraying; intent is not proven.
Unusual access / context shift	Review successful login after prior failures from a different IP/device, new device, or unusual time; correlate with known user baseline.	stu-311 failed from 203.0.113.42 on device unknown at 09:21:13; later succeeded from 192.0.2.99 on iOS at 11:47:09.	Review lead only. Could be legitimate; no baseline or user confirmation is supplied.
Rate-limit response	Alert when a rate-limit control blocks repeated authentication activity; confirm enforcement and record counters.	A blocked rate_limit event for stu-209 from 203.0.113.42 at 09:22:01 UTC.	Control response observed; determine scope and whether other accounts were protected.

Interpretation notes

• **Repeated failures are evidence of repeated unsuccessful attempts**, but do not establish whether a person, script, or benign misconfiguration generated them.

• **Multi-account failures from one source** are consistent with password spraying. The log does not show the attempted passwords, user-agent, request IDs, or the actor's intent.

• **Account-lock and rate-limit events** indicate defensive controls were recorded. The sample does not reveal configuration, thresholds, or whether every attempt was captured.

4. Incident timeline

UTC timestamp	Observed event	Evidence-based interpretation
09:00:12	stu-104 password_login success from 192.0.2.18; Android.	Successful authentication; no suspicious indicator is visible in this row alone.
09:03:44–09:04:18	Three password_login failures for stu-117 from 198.51.100.27; Windows.	Repeated failures meet the report's brute-force-like triage rule.
09:05:03	account_lock success for stu-117; server.	Account-lock action follows the failure burst by 45 seconds. The result label 'success' likely refers to the lock action succeeding, not a successful login.
09:21:10–09:21:22	Four password_login failures across stu-209, stu-311, stu-415, and stu-502 from 203.0.113.42; device unknown.	Rapid multi-account pattern consistent with password spraying.
09:22:01	rate_limit blocked for stu-209 from 203.0.113.42; server.	A rate-limit block is recorded 39 seconds after the last observed failure in the burst.
10:15:32	stu-104 refresh_token success from 192.0.2.18; Android.	Refresh-token event succeeded on the same IP/device as the earlier stu-104 login; not suspicious on its own.
11:47:09	stu-311 password_login success from 192.0.2.99; iOS.	Different IP/device from the earlier failure context. Validate with user/session records; this alone does not show a link to the prior attempts.

5. Evidence versus assumptions

Evidence directly present in CSV	Inference / assumption to validate
Three failed password logins for stu-117 from one IP in 34 seconds; an account_lock event follows.	May be brute-force activity. Could also be a user repeatedly entering an incorrect password; inspect broader context.
Four account IDs fail from 203.0.113.42 within 12 seconds; a rate-limit block follows.	Consistent with password spraying or automated probing. No password values or actor attribution are available.
stu-311 has a failure from 203.0.113.42 with device unknown, then a later success from 192.0.2.99 on iOS.	Could represent legitimate activity, a separate login, or a related event. The CSV does not establish a causal relationship.
A successful refresh_token event for stu-104 uses the same IP/device as the earlier successful login.	Appears consistent with continuity, but token validity, session binding, and device trust cannot be assessed from these columns.

6. Recommended containment and investigation

1. Preserve evidence. Retain the original CSV and export surrounding identity-provider, application, rate-limit, and account-lock logs. Preserve timestamps, request IDs, user-agent strings, and outcome reasons.

2. Validate account status. Check the lock event for stu-117 and confirm whether the user reported repeated failed logins. Avoid assuming the account was compromised.

3. Review the multi-account burst. Search the full log window around 09:21–09:22 UTC for the same source, related request identifiers, additional target accounts, and other control actions.

4. Verify stu-311's later login. Confirm with the account owner or approved support process whether the 11:47:09 UTC login from 192.0.2.99 on iOS was expected. Review session issuance and subsequent activity.

5. Apply proportionate controls. Use rate limiting, progressive delays, risk-based MFA, breached-password screening, and account-lockout controls that resist abuse without creating easy denial-of-service opportunities.

6. Improve alerting. Alert on failure bursts per account, failures across many accounts per source, distributed attempts across sources, and unusual successful logins after failure patterns. Tune thresholds to baseline.

7. Logging improvements

• Record event time in UTC with consistent precision and server-side ingestion time.

• Include normalized outcome codes and reason fields so 'success' on an account_lock action cannot be mistaken for login success.

• Capture request/correlation ID, authentication method, user-agent, device/session identifier, MFA outcome, and policy action where appropriate.

• Record rate-limit counter/threshold, lockout start and end, and whether the control was enforced or merely evaluated.

• Protect logs from tampering, restrict access, define retention, and monitor gaps or ingestion delays.

8. Limitations and conclusion

The supplied file has only 12 rows and six columns. It contains no usernames, attempted password values, user-agent strings, request IDs, MFA events, geolocation, known-good device baselines, or corroborating application logs. Consequently, this report identifies suspicious patterns and review priorities but cannot confirm compromise, attribute an actor, or prove that the two failure clusters are connected. The strongest defensible conclusion is that the sample shows a brute-force-like burst against stu-117 and a password-spraying-like burst against four accounts, with account-lock and rate-limit responses recorded.

Submission note: This PDF is based on the actual supplied CSV. To meet the task's single-link requirement, place this report and the original CSV together in a view-only folder or repository and submit that share link.